

CO250 Assignment 2

September 21–28

- [1] An art thief has broken into a museum and wants to steal as many art pieces as they can. However, they are on their own and can only carry a limited amount of weight. They want to steal art totaling as much value as they can. The following is a collection of art pieces that they can steal and their weights and their values.

Piece	1	2	3	4	5	6
Weight (kg)	10	15	7	8	5	3
Value (\$)	50	100	30	50	12	15

- [a] Formulate an integer program that models the thief's situation, assuming that they can only hold 30kg of art. Clearly explain your variables, objective function and constraints.

Answer:

let $x_i \in \{0, 1\}$ be a binary variable representing the choice of taking item i .

let $x = (x_1, x_2, x_3, x_4, x_5, x_6)$

Objective: $\max[50, 100, 30, 50, 12, 15]x$

subject to: $[10, 15, 7, 8, 5, 3]x \leq 30$

$x_i \in \{0, 1\} \quad \forall i \in [6].$

- [b] Suppose that in fact, the art thief has two bags, each of which can hold 15 kg of art. How do we need to change the previous integer program to model this change?

Answer:

let $x_i \in \{0, 1\}$ be a binary variable representing the choice of taking item i in sack 1.

let $y_i \in \{0, 1\}$ be a binary variable representing the choice of taking item i in sack 2.

let $x = (x_1, x_2, x_3, x_4, x_5, x_6)$ let $y = (y_1, y_2, y_3, y_4, y_5, y_6)$

Objective: $\max[50, 100, 30, 50, 12, 15](x + y)$

subject to: $[10, 15, 7, 8, 5, 3]x \leq 15$

$[10, 15, 7, 8, 5, 3]y \leq 15$

$x + y \leq \vec{1}$

$x_i \in \{0, 1\} \quad \forall i \in [6]$

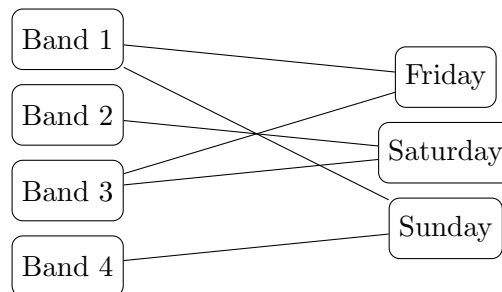
$y_i \in \{0, 1\} \quad \forall i \in [6]$

- [2] A music venue needs to schedule acts for the week. They operate Friday through Sunday, and each day, they schedule one act. There are 4 possible bands that they can schedule, but each band has only certain nights they can play. Below is a table that shows which bands are available on which nights.

	Friday	Saturday	Sunday
Band 1	✓		✓
Band 2		✓	
Band 3	✓	✓	
Band 4			✓

- [a] Draw the availability data as a bipartite graph. (A graph $G = (V, E)$ is bipartite if V can be partitioned into two parts A and B such that every edge e in E has exactly one end in A and one end in B .)

Answer:



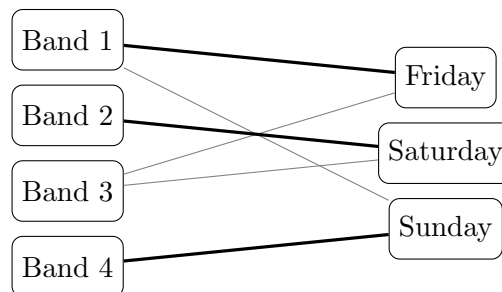
- [b] What does a valid scheduling correspond to in this bipartite graph? Find and draw a valid schedule for this music venue.

Answer:

A valid schedule corresponds to a matching of size 3 in the bipartite graph, where every day vertex is incident to exactly one selected edge.

For example, a valid schedule is:

$$\{(B_1, \text{Friday}), (B_2, \text{Saturday}), (B_4, \text{Sunday})\}.$$



- [c] Write down an integer program that solves this problem. The integer program should be of the form

$$\begin{aligned} \min \quad & c^\top x \\ \text{such that} \quad & Ax \leq b \\ & 1 \geq x \geq 0 \\ & x \in \mathbb{Z}^k, \end{aligned}$$

where A is a matrix and c is a vector, and k is an integer.

Answer:

Let

$$x = \begin{bmatrix} x_{1F} \\ x_{1Su} \\ x_{2Sa} \\ x_{3F} \\ x_{3Sa} \\ x_{4Su} \end{bmatrix},$$

where $x_{ij} = 1$ if the edge $\{i, j\}$ is chosen in the matching, and $x_{ij} = 0$ otherwise.

We want to maximize the number of selected edges. Since the required form uses a minimization problem, we equivalently minimize the negative of the number of selected edges. Thus,

$$c = \begin{bmatrix} -1 \\ -1 \\ -1 \\ -1 \\ -1 \\ -1 \end{bmatrix}.$$

For each vertex, at most one incident edge may be selected. Hence,

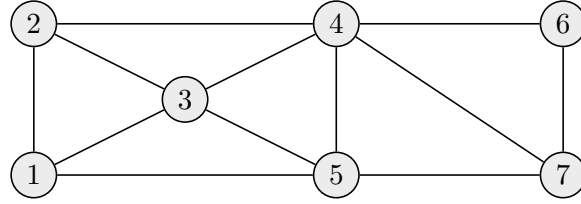
$$A = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}.$$

Therefore, the integer program is

$$\begin{aligned} \min \quad & c^\top x \\ \text{such that} \quad & Ax \leq b, \\ & 1 \geq x \geq 0, \\ & x \in \mathbb{Z}^6. \end{aligned}$$

- [3] We are given a graph G representing a network. A *triangle* in the graph is a set of three distinct nodes i, j, k such that all pairs ij, jk, ik are edges in G . Our goal is to pick as many triangles as possible so that no pair of triangles share an edge. Note that triangles are allowed to share nodes (triangles represent self-contained subnetworks that cannot share communication lines with other subnetworks). For the graph below the set of triangles $\{123, 345, 467\}$ would be a feasible solution; the set of triangles $\{123, 135\}$ would not be a feasible solution (the edge 13 is used by two triangles).

- [a] Formulate the above-mentioned problem as an IP for the following graph G .



Hint: First, list all the triangles in G . Then introduce a binary variable for each triangle. Then, for each edge ij in the graph, write a constraint that “says:” among the triangles which use this edge ij we may pick at most one of them.

Answer:

The triangles in G are

$$T = \{123, 234, 135, 345, 457, 467\}.$$

For each triangle $ijk \in T$, let

$$x_{ijk} = \begin{cases} 1, & \text{if triangle } ijk \text{ is selected,} \\ 0, & \text{otherwise.} \end{cases}$$

We want to select as many triangles as possible. Thus,

$$\begin{aligned} \max \quad & x_{123} + x_{234} + x_{135} + x_{345} + x_{457} + x_{467} \\ \text{subject to} \quad & x_{123} + x_{135} \leq 1, \\ & x_{123} + x_{234} \leq 1, \\ & x_{135} + x_{345} \leq 1, \\ & x_{234} + x_{345} \leq 1, \\ & x_{457} + x_{345} \leq 1, \\ & x_{457} + x_{467} \leq 1, \\ & x_{123}, x_{234}, x_{135}, x_{345}, x_{457}, x_{467} \in \{0, 1\}. \end{aligned}$$

- [b] Now, suppose we are given an arbitrary graph G with node set V and edge set E . Let T denote the set of triangles in G . Write down your IP model for this general set-up.

Answer:

For every triangle $t \in T$, let

$$x_t = \begin{cases} 1, & \text{if triangle } t \text{ is selected,} \\ 0, & \text{otherwise.} \end{cases}$$

We want to select as many triangles as possible, so the objective is

$$\max \sum_{t \in T} x_t.$$

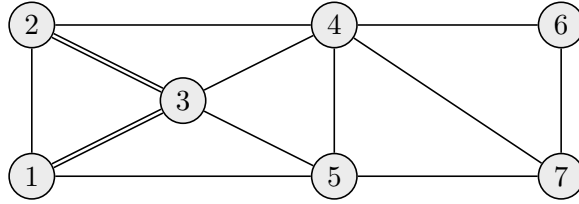
For each edge $e \in E$, at most one selected triangle may contain that edge. Thus, for every $e \in E$,

$$\sum_{\substack{t \in T \\ e \in t}} x_t \leq 1.$$

Therefore, the integer program is

$$\begin{aligned} & \max \quad \sum_{t \in T} x_t \\ & \text{subject to} \quad \sum_{\substack{t \in T \\ e \in t}} x_t \leq 1 \quad \forall e \in E, \\ & \quad \quad \quad x_t \in \{0, 1\} \quad \forall t \in T. \end{aligned}$$

- [c] Next consider a graph G whose edge set E is made up from two types of edges E_1 and E_2 . In these graphs the edges in E_1 have the same rule as before; but, the edges in E_2 can be used by at most two triangles (instead of at most one). For the graph below where $E_2 := \{13, 23\}$, formulate the new problem as an IP. Your formulation should be easily generalizable to any graph G .



Answer:

For every triangle $t \in T$, let $x_t \in \{0, 1\}$ indicate whether triangle t is selected.

$$\begin{aligned} & \max \quad \sum_{t \in T} x_t \\ & \text{subject to} \quad \sum_{\substack{t \in T \\ e \in t}} x_t \leq 1 \quad \forall e \in E_1, \\ & \quad \quad \quad \sum_{\substack{t \in T \\ e \in t}} x_t \leq 2 \quad \forall e \in E_2, \\ & \quad \quad \quad x_t \in \{0, 1\} \quad \forall t \in T. \end{aligned}$$